Student Patent Novelty Check

Provide more detail → better retrieval. Use keywords and filters to steer the search.

API: `http://127.0.0.1:8000`

LIVE · patentsview_live_semantic_anchor_gate (strict_anchors+keywords)

Backend health

Search summary

Copy this into your notes or for debugging.

Copy

Mode	patentsview_live_semantic_anchor_gate (strict_anchors+keywords)
Problem	We want to reduce patient readmissions for heart-failure patients after discharge by using a smartphone + wearable system that continuously…
Novelty	Uses a multi-modal transformer combining wearable time-series + patient self-reports + medication adherence to predict a…
Technologies	AI/ML, Mobile, NLP, IoT, Cloud
Keywords	heart failure, readmission, wearable, remote patient monitoring, decompensation, SpO2, respiration rate, heart rate variability, multimodal, early warning
Exclude	pacemaker, defibrillator, stent, imaging, ultrasound, catheter, printer, messaging
Assignee	Philips
Years	2015 → 2023
CPC refine filters	G09B

Describe your idea

More input usually improves relevance — especially keywords + exclusions.

What problem are you solving?

We want to reduce patient readmissions for heart-failure patients after discharge by using a smartphone + wearable system that continuously monitors vitals (heart rate variability, respiration, SpO2, activity), detects early decompensation using ML, and triggers personalized interventions. The system should work with noisy real-world signals, handle missing data, and provide explanations for alerts to clinicians.

What does your idea do?

- Automates process
- Analyzes data
- Reduces time
- Improves UX
- Hardware control

Primary domain

MedTech

Tip: Domain doesn't change semantic similarity much; keywords help more.

Key technologies

- AI/ML
- Cloud
- Mobile
- IoT
- Networking
- NLP
- Database

What's novel about your idea? (optional)

Uses a multi-modal transformer combining wearable time-series + patient self-reports + medication a

Power inputs

Advanced

These steer the search hard. Comma or new-line separated.

Must-include keywords

heart rate variability
multimodal
early warning

Exclude keywords

catheter
printer
messaging

Assignee (optional)

Philips

Max results

10

Year from

Year to

2015

2023

Do you want results to show a short abstract preview?

Note: This will only display if the backend returns `abstract` / `abstract_snippet` in results.

Find Similar Patents

Results

CPC used

G16H

Idea about: We want to reduce patient readmissions for heart-failure patients after discharge by using a smartphone + wearable system that continuously monitors vitals (heart rate variability, respiration, SpO2, activity), detects early decompensation using ML, and triggers personalized interventions. The system should work with noisy real-world signals, handle missing data, and provide explanations for alerts to clinicians. · Domain: MedTech · Mode: patentsview_live_semantic_anchor_gate (strict_anchors+keywords)

LIVE · patentsview_live_semantic_anchor_gate (strict_anchors+keywords) **7 results**

1 CPC **Top match: 52%**

Refine by CPC

CPC = patent category code. Clicking re-runs `/search` with `cpc_filters`.

Clear

G09B 3
Technical category

G16H 3
Digital health / healthcare informatics

A61M 1
Devices for introducing media into the body

Active CPC filters: **G09B**

Sort Relevance

Min similarity 0%

Reset

#1

Wearable device for sleep assistance

Similarity

US10478589 · 2019 · KONINIKLIJKE PHILIPS N.V.

52% Top match

A61M **Low confidence**

Show abstract preview

A wearable device (105) for sleep assistance includes at least one sensor (115) to detect one or several vital signs of a user of the wearable device, a memory (155) to store a set of sleep aid techniques, and an interfa…

Hide why similar

- Also automates part of the workflow you described.
- Uses data analysis to reach decisions, similar to your idea.
- Involves controlling hardware or equipment, like your use case.
- Falls under a similar domain (MedTech) in the CPC scheme.
- Mentions technologies that overlap with yours: AI/ML, Mobile, NLP, IoT, Cloud
- Classified in CPC area A61M, which is related to your idea's focus.

Feedback

Helps us learn what "relevant" means for students.

👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

Submit feedback

#2

Cardiopulmonary resuscitation guidance method, computer program product and system

Similarity **51%**

US10332423 · 2019 · KONINIKLIJKE PHILIPS N.V.

G09B **Low confidence**

Show abstract preview

A hands-free cardiopulmonary resuscitation (CPR) guidance method (40) for a system including a head-mountable computing device (100) comprising a processor (110) and at least one display module (106, 106′) arranged to be…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#3

Method and apparatus for exercise monitoring combining exercise monitoring and visual data with wireless wearable devices

Similarity **46%**

US9943723 · 2018 · KONINIKLIJKE PHILIPS N.V.

G16H Low confidence

Show abstract preview

Embodiments of the invention provide a method and apparatus for a wireless exercise monitoring system for interactively monitoring an aspect of exercise, sports, or fitness utilizing a wearable device, such as a watch, e…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#4

Method and apparatus for exercise monitoring combining exercise monitoring and visual data with wireless devices

Similarity **45%**

US10632342 · 2020 · KONINIKLIJKE PHILIPS N.V.

G16H Low confidence

Show abstract preview

Embodiments of the invention provide a method and apparatus for a wireless exercise monitoring system for interactively monitoring an aspect of exercise, sports, or fitness utilizing a wearable device, such as a watch, e…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#5

Software application for a portable device for CPR guidance using augmented reality

Similarity **45%**

US10134307 · 2018 · KONINIKLIJKE PHILIPS N.V.

G09B Low confidence

Show abstract preview

A method for guiding a user in performing a Cardio-Pulmonary Resuscitation (CPR) procedure on a patient involving a user portable device comprising a camera, and a display. The method includes receiving a video of the pa…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#6

System and method to monitor and titrate treatment for high altitude-induced central sleep apnea (CSA)

Similarity
42%

US11837106 · 2023 · KONINIKLIJKE PHILIPS N.V.

G09B Low confidence

Show abstract preview

An apparatus and method for employing data from the person and from the environment where the person is situated facilitate the detection and prediction of severity of high altitude-induced Sleep Disordered Breathing (SD…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 Relevant 👎 Not relevant

Optional: what did you expect to see instead?

Submit feedback

#7

System and method for providing a patient with personalized advice

Similarity
37%

US10679745 · 2020 · KONINIKLIJKE PHILIPS N.V.

G16H Low confidence

Show abstract preview

The present invention relates to an apparatus (10) for providing a patient (12) with personalized advice (56) relating to a use of a patient interface (18) for providing a flow of breathable gas and/or to a use of a pres…

Show why similar

Feedback

Helps us learn what “relevant” means for students.

👍 **Relevant** 👎 **Not relevant**

Optional: what did you expect to see instead?

Submit feedback

Tip: If you’re getting “printer / messaging” type patents, add those words to **Exclude keywords**.